Chemotherapy: Principles & Major Regimens

Mechanisms of Action, Drug Classes, Resistance, and Toxicity Management

Doc #03 of 15	**Domain:** Oncology & Cancer Treatment	**Audience:** Medical Professionals

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1. Pharmacological Principles

Cytotoxic chemotherapy exploits the higher proliferative rate of neoplastic cells relative to most normal tissues. Cell-kill follows first-order kinetics (log-kill hypothesis): a given dose kills a fixed fraction — not a fixed number — of cells. This underpins the rationale for combination regimens (non-overlapping toxicity, complementary mechanisms) and dose-dense scheduling. Pharmacokinetic parameters critical to efficacy include AUC, Cmax, and time above threshold concentration (schedule-dependent vs. concentration-dependent agents).

2. Major Drug Classes

Class	Example Agents	Mechanism	Phase Specificity
Alkylating agents	Cyclophosphamide, Carboplatin, Cisplatin	DNA crosslinking / adduct formation	Cell-cycle non-specific
Antimetabolites	5-FU, Gemcitabine, Methotrexate, Pemetrexed	Nucleotide synthesis inhibition	S-phase specific
Topoisomerase I inh.	Irinotecan, Topotecan	Stabilise topo I–DNA cleavage complex	S-phase specific
Topoisomerase II inh.	Doxorubicin, Etoposide	Stabilise topo II–DNA complex; intercalation	G2/M-phase

Taxanes	Paclitaxel, Docetaxel, Nab-paclitaxel	Microtubule stabilisation (prevents depolymerisation)	M-phase specific
Vinca alkaloids	Vincristine, Vinorelbine	Microtubule destabilisation (inhibit polymerisation)	M-phase specific
Antitumour antibiotics	Bleomycin, Mitomycin C	DNA strand breaks / crosslinks	Variable

3. Common Combination Regimens

Acronym	Agents	Indication	Cycle
FOLFOX	5-FU + Leucovorin + Oxaliplatin	CRC (adj. / metastatic)	q2w
FOLFIRI	5-FU + Leucovorin + Irinotecan	CRC (2nd line)	q2w
AC-T	Doxorubicin + Cyclophosphamide → Paclitaxel	Breast (adj.)	q3w x4 → q3w x4
ABVD	Doxorubicin + Bleomycin + Vinblastine + DTIC	Hodgkin lymphoma	q4w
BEP	Bleomycin + Etoposide + Cisplatin	Germ cell tumours	q3w x3-4
CHOP	Cyclophosphamide + Doxorubicin + Vincristine + Prednisone	DLBCL	q3w x6
Gem-Carbo	Gemcitabine + Carboplatin	NSCLC, Urothelial	q3w
GnP	Gemcitabine + Nab-paclitaxel	Pancreatic adenocarcinoma	q4w

4. Drug Resistance Mechanisms

- Reduced drug influx — downregulation of transporters (e.g., OCT1 for imatinib, RFC for methotrexate)
- Enhanced drug efflux — P-glycoprotein (MDR1/ABCB1) overexpression — multi-drug resistance
- Increased DNA repair capacity — ERCC1 overexpression reducing platinum efficacy
- Target alteration — BCR-ABL1 kinase domain mutations (T315I 'gatekeeper') conferring TKI resistance
- Bypass signalling — MET amplification rescuing EGFR-blocked pathways
- Epithelial-mesenchymal transition (EMT) — reduced apoptotic sensitivity
- Tumour microenvironment — hypoxia, drug sequestration, stromal barrier effects

5. Toxicity Management

Toxicity	Common Causative Agents	Management Approach	Grading (CTCAE)
Febrile neutropenia	Anthracyclines, taxanes	G-CSF (filgrastim/pegfilgrastim), IV antibiotics	Grade 3-4 = hospitalise

Nausea/Vomiting	Cisplatin, anthracyclines	5-HT3 antagonist + NK1 antagonist + dex	Per Hesketh emetic risk
Peripheral neuropathy	Oxaliplatin, paclitaxel	Dose reduction; duloxetine (limited evidence)	Grade 3 = dose modify
Cardiotoxicity	Anthracyclines	Lifetime dose cap (doxorubicin <450 mg/m2); echo monitoring	LVEF monitoring
Nephrotoxicity	Cisplatin	Pre/post IV hydration; avoid NSAIDs, nephrotoxins	Cr clearance monitoring
Mucositis	5-FU, methotrexate	Cryotherapy, oral hygiene, palifermin	Grade 3-4 = treatment hold

■ DOSE MODIFICATION: Hepatic or renal impairment requires dose adjustment for most cytotoxics. Consult institutional protocols and current SmPC/prescribing information before administering.